Printed Pages: 01 **Sub Code: RPH -301**
Paper Id: 233048 **Roll No.** | | | | | | | | | | |

B. PHARM
(SEM III) THEORY EXAMINATION 2022-23
HETEROCYCLIC & BIOORGANIC CHEMISTRY

Time: 3 Hours ***Total Marks: 70***

Note: Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief. 2 x 7 = 14

(a) Give the chemical structure of any two disaccharides.
(b) Draw the structure of pyrrole and pyrazole.
(c) Define amino with example.
(d) Differentiate between DNA and RNA.
(e) Write the formula to determine iodine value.
(f) Briefly classify vitamins.
(g) Which hetero atom is present in thiazole?

SECTION B

2. Attempt any *three* of the following: 7 x 3 = 21

(a) Illustrate the structure elucidation of retinol.
(b) Describe the structure of nucleic acid.
(c) Write a short note on analysis of fat and oil.
(d) Explain solid phase peptide synthesis.
(e) Elaborate the evidences of open chain structure of glucose.

SECTION C

3. Attempt any *one* part of the following: 7 x 1 = 7

(a) Write down the methods of preparation of furan and thiophene.
(b) What are carbohydrates? Classify them with proper examples.

4. Attempt any *one* part of the following: 7 x 1 = 7

(a) Differentiate between oils and fats. How will you determine the RM value of fats and oils?
(b) Which compound is more basic, pyridine or pyrimidine?

5. Attempt any *one* part of the following: 7 x 1 = 7

(a) Define the term polymerization along with important reactions of polymers.
(b) Define and classify vitamins. Discuss the structure of niacin.

6. Attempt any *one* part of the following: 7 x 1 = 7

(a) Write a note on end group analysis of proteins.
(b) Explain genetic codes.

7. Attempt any *one* part of the following: 7 x 1 = 7

(a) How to separate and purify proteins?
(b) Explain the IUPAC nomenclature of heterocyclic compounds with proper examples.

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